Sahaj Bhadja

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Summary

Software Engineer with 2+ years delivering **system emulation, automated testing, software configuration, deployment, and production support** across distributed applications, simulation environments, and customer-facing implementation workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Computer Networks, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Computer Architecture

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reworked asynchronous FastAPI orchestration with explicit job states and failure isolation, processing 1K+ concurrent jobs at 3x throughput while sustaining **software reliability, runtime monitoring, configuration validation, and post-implementation support**.
- Optimized indexed Azure SQL queries and row-level isolation for 500+ users, cutting latency from 1.2 seconds to 350 milliseconds through **performance testing, failure diagnosis, status reporting, and permanent corrective actions**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **automated tests, functional validation, debugging, and fault isolation**.
- Delivered React and Flask workflows supporting 150+ instructors and 10K+ daily requests while coordinating **software demonstrations, test feedback, deployment verification, and release support** across development and user teams.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving offline reinforcement learning, improving planning 18% while processing 100K+ simulated trajectories through reproducible **equipment-like emulation, scenario testing, and performance validation pipelines**.
- Designed spatial-temporal encoders combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **geometric-layout interpretation, simulated control behavior, systematic test analysis, and documented results**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed cross-platform APIs and data pipelines using object-oriented patterns, AWS Glue, S3, and caching, improving response times 30% while reducing technical debt 20% across integrated software components.
- Automated 20+ unit and integration suites across Windows, Linux, and macOS using PyTest, Mockito, Jenkins, and virtual machines, strengthening **regression testing, installation validation, defect resolution, and commissioning readiness**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight modular services for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, component interfaces, and real-time operational status updates** across locations.
- Automated Docker and Jenkins deployments, reducing release time 50% while using Sentry, Azure Monitor, and Kibana for **startup verification, incident diagnosis, on-call support, and reliable multi-environment commissioning**.

Skills

Commissioning: Software Installation, Configuration, Onsite Commissioning, Startup Support, Post-Implementation Support, Status Reporting
Testing & Emulation: Equipment Emulation, Functional Testing, Unit Testing, Integration Testing, Regression Testing, Automated Testing, Acceptance Criteria
Software: C#, Java, Python, C++, JavaScript, REST APIs, WebSockets, SQL, Object-Oriented Design, Multithreading
Systems & Automation: WCS, WMS, PLC Interfaces, Material Handling, Conveyors, Shuttles, Robotics, Docker, Jenkins, CI/CD, Git, Linux
Operations: Troubleshooting, Root-Cause Analysis, Monitoring, On-Call Support, Technical Documentation, Customer Communication, Microsoft Excel, PowerPoint

Projects

Warehouse Equipment Emulation Lab | *C#, Python, WCS, PLC Interfaces, Docker*

- Built a **warehouse equipment emulation model** representing conveyors, shuttles, sensors, queues, and fault states, enabling repeatable WCS functional tests, acceptance scenarios, team demonstrations, and comprehensive pre-site software validation.

Commissioning Test Console | *C#, .NET, SQL, Automated Testing, Power BI*

- Designed a **commissioning validation and reporting console** that executes scripted functional tests, records subsystem results, tracks defects and readiness, visualizes project status, and packages evidence for onsite handoffs.

CPU Scheduling Simulator | *C++, React, JavaScript, Algorithms* | Live

- Implemented performance-sensitive C++ algorithms for FCFS, SJF, SRTF, round-robin, priority, and multilevel queues, validating correctness and visualizing runtime behavior across interactive workloads through a browser-based scheduling simulator.